

A Packing of 16 Four-Spaces in $\mathbb{R}^{10}$

Limin Ge*

October 9, 2026

Abstract

We give 16 4-dimensional subspaces of $\mathbb{R}^{10}$ whose smallest squared chordal distance is at least 2.559634. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.549645, obtained by deleting 1 of the 17 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 16 subspaces of dimension 4 in $\mathbb{R}^{10}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.559634 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 10$, $n = 4$. The cell $N = 16$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.549645, obtained by deleting 1 of the 17 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(10 - 4)/10 \cdot 16/15 = 2.560000$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m10n4N16.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 63 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
341284597023560935224182489292759059120346975453015762721492410647936523
539016680999033032319645601783343938967717696660425304359765271354269512
73810980399085775236247055691626727801010321646
```

denominator:

```
133333333333396227154589427736178316392626759331203029113967829961795448
267285425813572667632085975121719734439912320744640480279313228210512151
09725152651475764622258951742743166415326357981
```

B Coordinates

16 blocks of 10 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m10n4N16.txt`.

```
371580165536 189168251742 -768655211553 -266052353042342779366 535175971792 -6034620174 4500242884 293400408904 -200419267589 -32920788257 -375059909715
-330530405104 -32795829444 -219603182564 543248630232 1136774570 -533162934134 256012490608 8719171781172495296328 256265484702 -447591741170 -78607819920
-250013066078 212068103529 -249877595729 -144335703113896273731 242019211403 -145093877324 25057783063687276391606 383575168412 225064978899 -120523313101
565306095636 -16484960315 141411038233 48699360368286222376508 227066007393 13149377481 59353587651 -256906386601 503710921092 -35516125406 326394143619
-323415880601 515732294444 70700296884 20964107577 456111214244 -98473159200 670634071904 -4038891262386728772298 29158071875 -557103423063 35330114319
371939214299 244199522400 453981833727 -3601164014254165238962 324618491335 173879925311 -31451983646224124300033 351678873076 381994104274 -311167459206
-133748648240 -496713994153 37427701094 31337045526 131922283108 25100721144 472590269047 -2502302628531370148638 490246960360 -39990547505 47780667951
257858585696 341849543763 -114067719385 47649800708422818421112 -162077037264 -126155284947 -59356711522066763698 301790159218 -614089673557 -323899845177
-25812934930 -38563079390 -94048690415 -115779774230159910009319 14928068523 13497462556 -64500763215 -111715280240 196151178220 123433996091 341391718796
-206440420506 476664401575 210124349998 67385988046174212086287 -128309553907 -252853705980 17015800284207945687181 -228508360525 227452416470 -252711855744
-339978967377 -8335991549 -322682916780 -134581655059239600389 901353531291 148022364520 -21969339341675597470953 -214624089079 -162598762657 474106332086
499009412628 -334510688509 -436144621307 -23294607865994851696 -174296536507 -165707178 -334652730691440442295420 -80267912168 -140308084567 -518155464710
29819663356 171753729653 -198949553653 711219977108265434727175 226668574272 -740752575865 2657534774254479056031 171718178472 -374884264310 217163155949
-406616650574 257053723935 -386605579328 150678724209874405283 -125121597565 1828594070 -30569596298 -328344579740 -564473832439 -450041625540 241621244182
-626445992141 -372174037301 27751448589 260507683229323982327 -207096725384 226222295396 -1236612043087746202371 -141426134876 326360604938 556060106323
-117989826488 133362093837 -560338888443 -427988294326568603770 60615324159 273534652943 612175073683111551684263 -502669140940 364483634880 65817533174
-95858038156 540813588639 191948420313 -10201885765245521834660 -212926737912 -9386828978 -1840806696220255898739 -252607538736 -345871833411 -409442349522
-42221585711 -423555631193 -129930900322 348477272282850706505 -151553985406 -357205317097 -13338004420218683529 -373664731057 38411759538 82314879806
167416567597 395226691135 -143537680272 88244415484296919224716 109640657519 -511486995505 2014356340477999458755 246408745714 165282414265 436529427572
154342741633 89862895612 -357112996515 340602297438184391349970 407111474120 36324834209 -820069746709718659699123 -247533207233 -99782473650 207923758562
-133896123511 186729583773 691675211339 -1300174310929387311378 75148101239 451378098790 265068187557 368498865530 223763212162 -337515633298 284184913894
192399681678 402424567295 -267125787853 1964057328961379840441 -265419871089 528139711899 -7274641912476003510294 -60899172147 381143476203 -303239963932
-358534334117 -341162391484 -367156218693 30613836963014217381 -582795660650 21964476800 -2054621535353938885332 398725129400 -230638556506 256407944942
115959738228 -119960721247 8265521542 148727274215 -550805480093 -298966249024 -161050761052 -30985972730896296382 -271513920426 130774710030 -123627104688
39480953494 377225921287 -519934869478 -5927564959424455396396 -440184314424 40194605103 -7045049456642571029458 369834163944 178371886582 -560748796360
-106304464069 -313816987534 -449849807973 31323072330443940446 -234958210414 -314115700373 1284101303981824754293 -284176292536 -482001222889 373701019117
473443391949 -486461545516 39570298477 -43928375133871424563985 -203456389656 -388651765580 -199478028202294603817 323280632005 -112166350094 283112143042
416424034277 328472530938 164675501865 -714461137 44777804286 121121575294 -13666166391 163173087278 292251550200 -429687782090 -59958022544 -305456203578
412351316375 521113710110 -116955305419 40681254500665386467607 -353823692875 309150985003 144966411581129018159618 116760181644 -727425764806 -525158903428
471815691111 -290463503705 -25276953262 14400982990468815845437 152212197187 -78040465737 159338497392632810199 -36247324970 -59765962505 56496370714
289119454376 -164740920854 -222920579671 -1629624257259204145432 -219420071669 338719687586 680378750734585944260 272823738754 -207834779690 102926330718
297977105204 148102547133 115410030677 60948715490647491285326 -157209995093 -310687784574 -2654252998128259400538 -202905006935 -271054162286 -42966490406
129984340657 -614690847517 -265265880184 -37796815520787630903 -288390951728 -246377117197 -472080302203820223483 23588405507 289080417828 -388787901251
413056540212 -145918623898 80047338663 25268823990661862655571 296601668364 540058111514 82204275111 49620894466 460345527911 -211225195422 -221706480995
85083068011 28247959102 437683572049 -479241472640 143501662744 -741061809612 176307418505 30622056101779968783054 209537153962 172208677316 466291250410
-39277567718 -386317481684 -247989502603 -1846863595840097788 -48996490749 434511756986 -75300362031464552039899 -26953458354 -222891662137 367309370351
456904379481 209669678151 -400865176205 125509757701918134405 -232976653696 -561406425485 3104663569220334713132 468875746812 621983765416 35927238196
-261975640404 397098054576 -110596191301 446887429086131651249 110264155408 150899903935 277512829920291847073244 -123355214591 419540967738 68671964043
490428308217 -32483429541 541060495966 -1768450050290374401202 19429642589 391997307213 740784705632 6429044089 -242681745394 29288085682 -145899046311
-336768944835 -446133503437 376134027190 473163111928082278683 97443521466 341530010278 -23122398205804434288588 -614751059638 409398222100 30474545332
403039396139 42433363004 361210471815 -48954634965 512246798081 -418193066248 164289730573 80251171816181517250684 113945161222 -62600225425 61709668467
136719860854 -522080293704 -557565260053 955341324125835147925 292193381890 349109056878 122732656371 244997266669 -54319701964 -239202831834 -194121001099
-475470379187 336309276841 -399105846208 -18554616388441205182 -202453810666 209728048076 -293485576833708214997 -298849127792 425554539042 -35963959475
168850114418 393997532287 -295815757476 53684746821482293894414 -730944683054 347862740284 -897931763971809906048 234715248238 -204459729930 172671839604
-120012546930 -167865008246 -347844517330 24431491233329964072 -135301092556 -220260634176 -6038605536659815089226 -293548091892 211030367704 -535369947604
656020704877 242168924319 -279678691336 -241243228476252036211 -264299961892 -171376985831 324074267839267421972 225197609675 201923984629 -222496735444
-319568984075 33302454631 95751400960 -314813154034404963883591 -5877771658 413319915679 201692267445385547825389 171713053339 -17998612812 -48007712891
1108671873 218431499378 -22467368776 -88964956174812277523429 155055617830 112847544359 296467863094497983311015 -17666893142 319970687417 682360087712
-129577459923 45157494931 303854772329 6640171269475102900568 269600288764 403586429333 -114594297467377800502213 485270442757 -429582844041 -182078232042
22357164389 -565339340433 103144181844 -3896961154 179839541222 -397447060791 238136295695 -45629395353238892057755 -290122105741 -373650842363 -13446752948
596720359792 -310821734420 -508027708677 -1822659566836781236 -103767176045 -118166198470 37527835802370814662720 -228104547970 -318227419456 359136104072
278686213637 -247438973760 182694778634 247414401830630748999 174135695472 -560173601652 27713383236972752133001 563228275167 404551836258 50330731670
117949062302 -21490547689 -328350387853 -20476386068669974131 -10032724486 147635398985 696323355975
-299694725008 9847625616 -184197607303 -72639734093373353904 773461906660 214156173205 -94456411260
```